POWER: USB-C 5V (D101) / LiPo (Q101) auto power path -> VSYS -> U6 LP5907 3.0 V -> +3V0_A (ADS1298 AVDD)
and U9 AP2112K 3.3 V -> +3V3 (ESP32-S3, ADS1298 DVDD). U6 EN = +3V3, so the AFE is off whenever the MCU rail is off.

B5819W
D101
VBUS
VSYS
#FLG07
PWR_FLAG
VSYS
AO3401A
Q101
LP5907MFX-3.0
U6
VSYS
+3V3
IN
EN
OUT
GND
+3V0_A
AGND
C101
1UF
C102
100NF
R108
100K
VBUS
VBAT
VSYS
C107
10UF
TEST POINTS
+3V0_A
+3V3
AGND
VSYS
RLD_OUT
AGND
LiPo pads (protected)
VBAT
AGND
LiPo: J3 = two solder pads for a protected LiPo. U8 MCP73831 charges it from USB at 100 mA (R_PROG 10k).
C120
100NF
C119
100NF
C118
10UF
#FLG02
PWR_FLAG
AGND
ADS1298IPAGR
U1
C121
1UF
C122
100NF
RLD: 1M || 1.5nF feedback, 330k series to the REF electrode(s) limits DC fault current to <10 uA.
R101
1M
C111
1.5nF
R102
330K
RLD_OUT
RLD_INV
RLD_ELEC
INPUT RC: 22k series (limits fault and plug-insertion current) + 100pF COG to AGND, ~72 kHz.
Kept small so high-impedance dry electrodes do not turn mains common-mode into differential hum.
ADS1298 oversamples at 512 kHz, so this is the only anti-alias filter needed.
CH1 (TRS)
J1
RLD_ELEC
ELEC1_N
ELEC1_P
ELEC1_P
ELEC1_N
ELEC2_P
ELEC2_N
ELEC3_P
ELEC3_N
R201 220k
C200 100pF C0G
CH1_P
CH1_N
CH2_P
CH2_N
CH3_P
CH3_N
AGND
CH1_P16
CH1_N15
CH2_P14
CH2_N13
CH3_P12
CH3_N11
IN1P
IN1N
IN2P
IN2N
IN3P
IN3N
IN4P
IN4N
IN5P
IN5N
IN6P
IN6N
IN7P
IN7N
IN8P
IN8N
VCAP1 28
VCAP2 30
VCAP3 55
VCAP4 26
VREFP 24
AGND 25
AVDD 19
AVDD1 54
DVDD 48
+3V0_A
+3V3
WCT 64
RLDOUT 63
RLDIN 62
RLDREF 60
TESTP_PACE_OUT1 17
TESTP_PACE_OUT2 18
GPIO1 42
GPIO2 43
GPIO3 44
GPIO4 45
AGND 41
DAISY_IN 40
SCLK 34
DIN 35
CS 39
DOUT 33
CLK 37
CLKSEL 52
START 38
DRDY 47
PWDN 36
RESET 46
RESV1 31
AGND 29
AGND 53
AGND 32
RLD_OUT
RLD_ELEC
RLD_INV
+3V3
ADS_START
ADS_DRDY
ADS_PWDN
ADS_RESET
SPI_SCK
SPI_MOSI
ADS_CS
SPI_MISO
AGND
AGND
AGND
SAFETY: with electrodes on the body and USB-C plugged in, run the laptop on battery,
unplugged from mains, or use a simulated input. Charge while worn only from
an isolated (Class II) charger.
LAYOUT: keep U1 and its supply away from the XIAO antenna.
ELECTRODES: J1 TRS for CH1 (T +, R -, S REF).
J6 1 mm FPC carries CH1-CH3 for a flex array or breakout,
in parallel with J1 for CH1: use one or the other.
All REF pins share RLD_ELEC. J6 1/5/9 + MP = AGND guards.
FLEX ARRAY CH1-3 (FPC)
J6
AGND 1
ELEC1_P 2
ELEC1_N 3
RLD_ELEC 4
AGND 5
ELEC2_P 6
ELEC2_N 7
RLD_ELEC 8
AGND 9
ELEC3_P 10
ELEC3_N 11
RLD_ELEC 12
AGND_MP
C112
10UF
C113
100NF
VREFP
C114
22UF
C115
1UF
C116
1UF
C117
1UF
VCAP1
VCAP2
VCAP3
VCAP4
ADS1298: CH1-CH3 used, CH4-CH8 and TESTP tied to AVDD per datasheet (power down in firmware).
1, 2 or 3 channels by mode, no rework. HPF/LPF in firmware: capture ~0.5 Hz up at 4 kSPS+.
Lead-off detection and RLD channel selection are firmware settings.
Internal clock (CLKSEL high), internal 2.4 V reference.
R103
10k
R105
10k
R106
10k
ADS_CS
ADS_RESET
ADS_PWDN
MCU: ESP32-S3-MINI-1-N8 (PCB antenna, keep copper out of the antenna end). Native USB on IO19/IO20 through USBLC6 ESD to USB-C.
ADS1298 SPI: IO12 SCLK, IO13 DOUT (MISO), IO14 DIN (MOSI), IO15 CS, IO16 DRDY, IO17 START, IO18 RESET.
Flashing over USB uses the built-in USB-Serial-JTAG. If it ever stops enumerating: short TP10 (IO0) to GND, then tap TP9 (EN) to GND.
Charger U8 MCP73831 at 100 mA (R703 10k). 3.3 V rail U9 AP2112K from VSYS.
J7
USB-C
VBUS
SHIELD
GND
SBU1
SBU2
CC1
CC2
D-
D-
D+
D+
A4 VBUS
A5 USB_CC1
B5 USB_CC2
A7 USB_DN
B7 USB_DN
A6 USB_DP
B6 USB_DP
A8
B8
A1
AGND_SH
AGND
#FLG08
PWR_FLAG
VBUS
R701 5.1k
R702 5.1k
USB_CC1
USB_CC2
AGND
AGND
U7
USBLC6-2SC6
USB_DP
USB_DN
VBUS
AGND
U2
ESP32-S3-MINI-1-N8
3V3
EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
TXD0
RXD0
USB_D-
USB_D+
IO21
IO26
IO33
IO34
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
GND
ESP_EN
ESP_BOOT
SPI_SCK
SPI_MISO
SPI_MOSI
ADS_CS
ADS_DRDY
ADS_START
ADS_RESET
USB_DN
USB_DP
AGND
U8
MCP73831T-2ACI/OT
VBUS
VDD
VBAT
PROG
VSS
STAT
CHG_PROG
R704 10k
C108 4.7uF
C109 4.7uF
VBUS
VBAT
AGND
U9
AP2112K-3.3TRG1
VIN
VOUT
EN
GND
VSYS
+3V3
C110 1uF
C104 10uF
C105 100nF
R704 10k
C103 1uF
ESP_EN
TP9
TP10
EN (reset)
BOOT (IO0)
ESP_BOOT
Spec: emg_headband/hardware/emg_afe/README.md
1 to 3 channel surface EMG, ADS1298 + ESP32-S3-MINI-1, JLCPCB assembly
Sheet: /
File: emg_wearable_revA.kicad_sch
Title: EMG wearable AFE
Size: A2
Date: 2026-10-11
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1